

Diabetes Risk Factor Analysis Using SQL and Tableau

Project Overview

Diabetes is a growing public health concern, and identifying populations at higher risk is essential for improving prevention and early intervention strategies.

This project analyzes diabetes prevalence and associated risk factors using the **CDC Diabetes Health Indicators dataset**. The goal was to explore how demographic characteristics, health conditions, lifestyle behaviors, and healthcare access factors are associated with diabetes prevalence.

The analysis was performed using **SQL Server** for data preparation, cleaning, transformation, and exploratory data analysis (EDA), while **Tableau** was used to create interactive dashboards and communicate key findings.

Business Objective

The objective of this project was to answer:

- What proportion of the population has diabetes?
- Which demographic groups show higher diabetes prevalence?
- Which health conditions are strongly associated with diabetes?
- How do lifestyle behaviors relate to diabetes prevalence?
- How does healthcare access differ among individuals with and without diabetes?

Dataset Description

The dataset contains health indicators collected from individuals and includes information related to:

Demographic Factors: Age category, Gender, Education level, Income level

Health Indicators: Diabetes status, High blood pressure, High cholesterol, BMI, General health status, Heart disease history, Stroke history, Difficulty walking

Lifestyle Factors: Physical activity, Smoking status, Fruit consumption, Vegetable consumption, Alcohol consumption

Healthcare Access Factors: Healthcare coverage, Cost barriers preventing healthcare visits

The dataset contains both categorical and numerical variables, requiring preprocessing before analysis.

Data Preparation and Cleaning (SQL)

SQL Server was used to prepare the dataset for analysis.

Data Cleaning Steps

The following steps were performed:

1. Data Import and Initial Exploration

- Imported the raw dataset into SQL Server.
- Reviewed dataset structure and column characteristics.
- Checked data completeness and consistency.

2. Data Transformation

Binary variables were converted into meaningful analytical categories. This improved readability during analysis and visualization.

3. Categorical Variable Interpretation

Categorical variables were transformed into understandable groups.

Examples:

Age: Age codes were converted into age categories to improve interpretation.

Education: Education levels were interpreted according to their original scale:

- No formal education
- Elementary education
- Some high school
- High school graduate
- Some college/technical education
- College graduate

Income: Income categories were converted into readable income groups.

General Health

The General Health variable was originally coded on a scale of 1–5, where each number represented a specific health category.

- 1 = Excellent
- 2 = Very Good
- 3 = Good
- 4 = Fair
- 5 = Poor

4. BMI Categorization

BMI values were transformed into standard weight categories:

- Underweight
- Normal weight
- Overweight
- Obese

This allowed comparison of diabetes prevalence across weight groups.

5. Summary Table Creation

A summary table was created to store important analytical findings, including:

- Analysis category
- Factor evaluated
- Group compared
- Metric
- Value
- Insight

This supported documentation of major findings from the exploratory analysis.

Exploratory Data Analysis (EDA)

EDA was conducted to identify patterns and relationships between diabetes and potential risk factors. The main analytical measure used was:

Diabetes Prevalence (%): This was selected instead of diabetes counts because different categories contained different population sizes.

For example: A group may have more diabetes cases simply because it has more people. Prevalence measures the proportion of people affected within each category.

Key Findings

Overall Diabetes Distribution

- Diabetes represented approximately **13% of the population**.
- The majority of participants were classified as non-diabetic.

Demographic Insights:

1. Age

- Diabetes prevalence increased with age.
- Older age groups showed the highest diabetes prevalence.
- The highest prevalence was observed among older adults, particularly the 65–79 age group.

Insight:

Age is strongly associated with diabetes prevalence, with older populations experiencing greater diabetes burden.

2. Gender

- Males showed a higher diabetes prevalence compared with females.

Insight:

Diabetes prevalence differed slightly by gender, with males showing a higher proportion of diabetes cases.

3. Income

- Lower-income groups showed higher diabetes prevalence.
- Diabetes prevalence generally increased as income level decreased.

Insight:

Lower socioeconomic status may be associated with increased diabetes risk due to differences in healthcare access, lifestyle, and preventive care opportunities.

4. Education

- Lower education groups showed higher diabetes prevalence.

Insight:

Lower educational attainment was associated with increased diabetes prevalence, potentially due to differences in health awareness and access to health information.

Health Risk Factor Insights

1. High Blood Pressure

- Participants with high blood pressure showed substantially higher diabetes prevalence compared with those without high blood pressure.

Insight:

Hypertension and diabetes frequently occur together and represent important cardiovascular risk factors.

2. High Cholesterol

- Participants with high cholesterol showed higher diabetes prevalence compared with those without high cholesterol.

3. BMI

- Obese individuals showed considerably higher diabetes prevalence compared with normal-weight individuals.

Insight: Higher BMI, particularly obesity, was strongly associated with increased diabetes prevalence.

4. General Health

- Participants reporting poorer general health showed higher diabetes prevalence.

5. Heart Disease

- Participants with heart disease history showed higher diabetes prevalence compared with those without heart disease.

6. Stroke

- Participants with previous stroke history showed higher diabetes prevalence compared with those without stroke history.

7. Difficulty Walking

- Individuals reporting difficulty walking showed higher diabetes prevalence compared with those without difficulty walking.

Lifestyle and Healthcare Insights

1. Physical Activity

- Inactive participants showed higher diabetes prevalence compared with active participants.

Insight: Physical inactivity was associated with increased diabetes prevalence.

2. Smoking

- Smokers showed slightly higher diabetes prevalence compared with non-smokers.

3. Fruit and Vegetable Consumption

Individuals consuming fruits and vegetables generally showed lower diabetes prevalence compared with those who did not.

4. Alcohol Consumption

- Differences were observed between alcohol consumption groups, although the relationship requires careful interpretation because lifestyle behaviors may interact with other factors.

5. Healthcare Access

- Individuals experiencing healthcare access barriers due to cost showed higher diabetes prevalence.

Insight: Healthcare affordability may influence diabetes prevention, diagnosis, and management.

Tableau Dashboard Development

Four interactive dashboards were created.

Dashboard 1: Diabetes Overview

Purpose: To provide a high-level summary of diabetes prevalence.

Includes: Total participants, Total diabetes cases, Diabetes prevalence percentage, Average BMI, Diabetes distribution overview

Diabetes Distribution

Total Number of People

253,680

Diabetes Cases

35,346

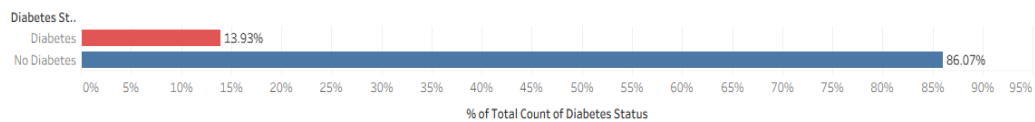
Diabetes Prevalence

13.93%

Average BMI

28.38

Diabetes Distribution Overview

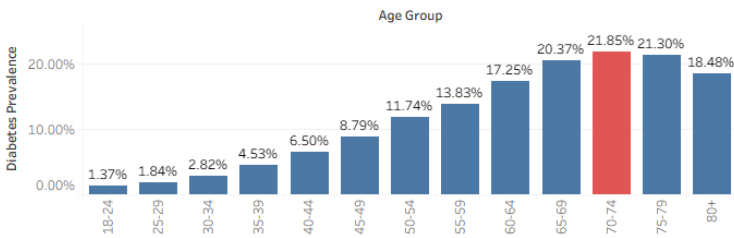


Dashboard 2: Demographics and Diabetes Prevalence

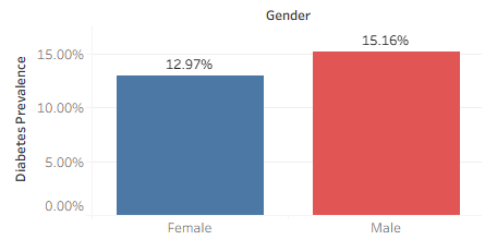
Purpose: To understand how diabetes prevalence varies across population groups.

Includes: Diabetes prevalence by age group, Diabetes prevalence by gender, Diabetes prevalence by income level, Diabetes prevalence by education level

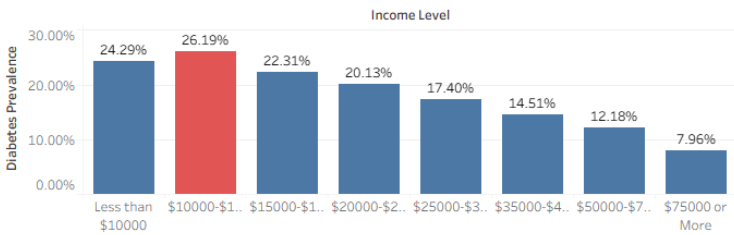
Diabetes Prevalence Across Age Groups



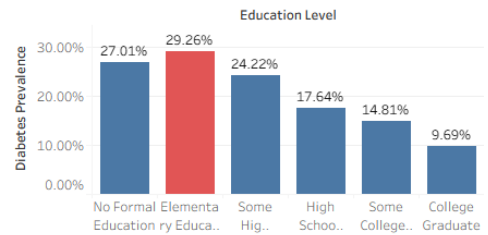
Diabetes Prevalence Across Gender Groups



Diabetes Prevalence Across Income Ranges



Diabetes Prevalence Across Educational Levels



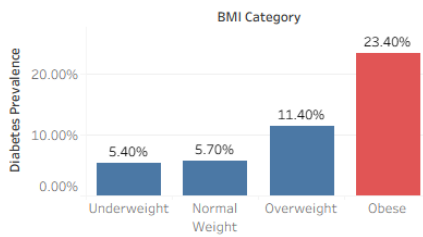
Dashboard 3: Health Risk Factors and Diabetes

Purpose: To explore relationships between diabetes and health conditions.

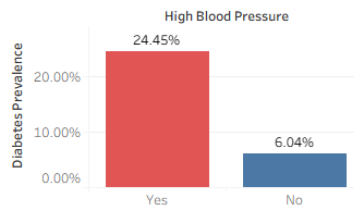
Includes:

- Diabetes prevalence by blood pressure status
- Diabetes prevalence by cholesterol status
- Diabetes prevalence by BMI category
- Diabetes prevalence by general health status
- Diabetes prevalence by heart disease status
- Diabetes prevalence by stroke status

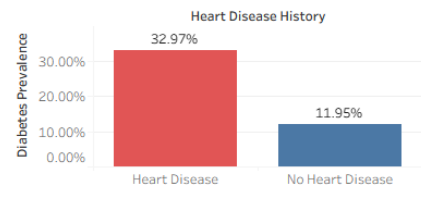
Diabetes Prevalence Across BMI Levels



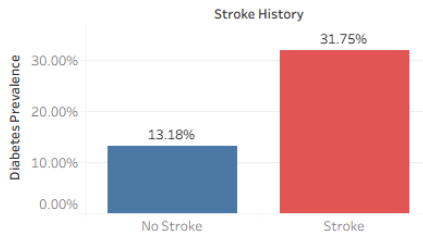
Diabetes Prevalence by High Blood Pressure Status



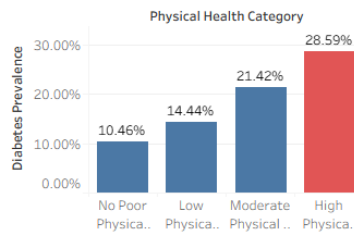
Diabetes Prevalence by Heart Disease Status



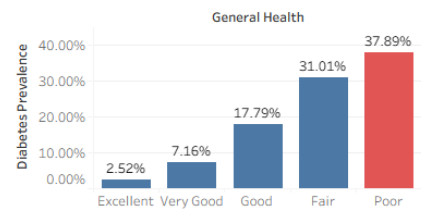
Diabetes Prevalence by Stroke Status



Diabetes vs Physical health



Diabetes Prevalence by General Health Status



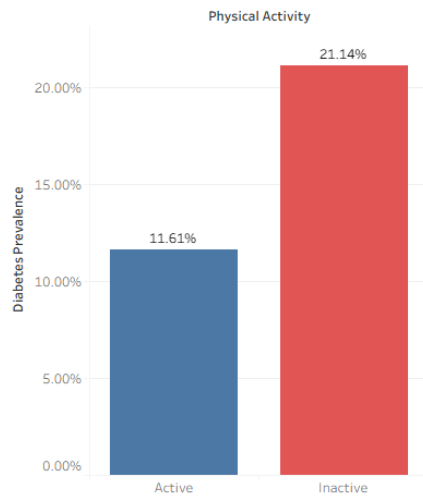
Dashboard 4: Lifestyle and Healthcare Factors

Purpose: To examine behavioral and healthcare-related factors associated with diabetes.

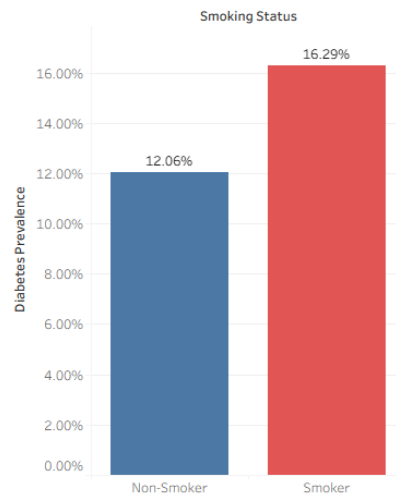
Includes:

- Diabetes prevalence by physical activity status
- Diabetes prevalence by smoking status
- Diabetes prevalence by fruit consumption
- Diabetes prevalence by vegetable consumption
- Diabetes prevalence by alcohol consumption
- Diabetes prevalence by healthcare coverage
- Diabetes prevalence by healthcare cost barriers

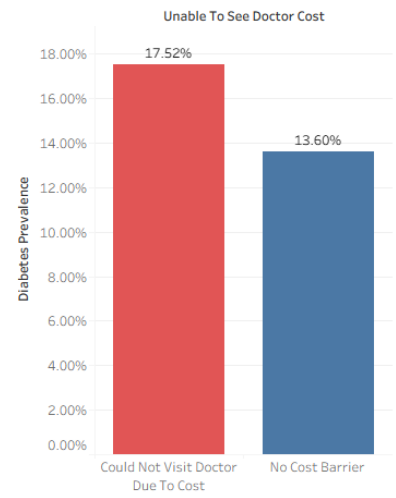
Diabetes Prevalence by Physical Activity Status



Diabetes Prevalence by Smoking Status



Diabetes Prevalence by Healthcare Cost Barrier



Tools Used

Data Analysis

- Microsoft SQL Server
- Microsoft SQL queries
- Exploratory Data Analysis

Data Visualization

- Tableau

Data Handling

- CSV datasets
- Data cleaning and transformation

Recommendations

1. Prioritize Screening Among High-Risk Groups

Healthcare programs should focus on:

- Older adults
- Individuals with cardiovascular conditions
- Individuals with obesity

2. Promote Healthy Lifestyle Behaviors

Encourage:

- Increased physical activity
- Healthy dietary habits
- Weight management

3. Improve Preventive Healthcare Access

Reducing cost barriers may improve:

- Early diagnosis
- Disease monitoring
- Diabetes management

4. Address Socioeconomic Differences

Diabetes prevention programs should consider differences related to:

- Income
- Education
- Healthcare accessibility

Conclusion

This project demonstrates how SQL and Tableau can be combined to transform raw healthcare data into meaningful insights.

Through data cleaning, exploratory analysis, and visualization, the project identified demographic, health, lifestyle, and healthcare factors associated with diabetes prevalence.

The analysis highlights the importance of targeted prevention strategies focused on high-risk populations and improved access to healthcare resources.